

You are viewing *main* version, which requires installation from source. If you'd like regular pip install, checkout the latest stable version ([v0.5.0-rc.0](#)).

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Safetensors

ML Safer For All

Safetensors

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Safetensors is a new simple format for storing tensors safely (as opposed to pickle) and that is still fast (zero-copy). Safetensors is really fast .

Installation

with pip:

```
pip install safetensors
```

with conda:

```
conda install -c conda-forge safetensors
```

Usage

Load tensors

```
from safetensors import safe_open

tensors = {}
with safe_open("model.safetensors", framework="pt", device=0) as f:
    for k in f.keys():
        tensors[k] = f.get_tensor(k)
```

Loading only part of the tensors (interesting when running on multiple GPU)

```
from safetensors import safe_open

tensors = {}
with safe_open("model.safetensors", framework="pt", device=0) as f:
    tensor_slice = f.get_slice("embedding")
    vocab_size, hidden_dim = tensor_slice.get_shape()
    tensor = tensor_slice[:, :hidden_dim]
```

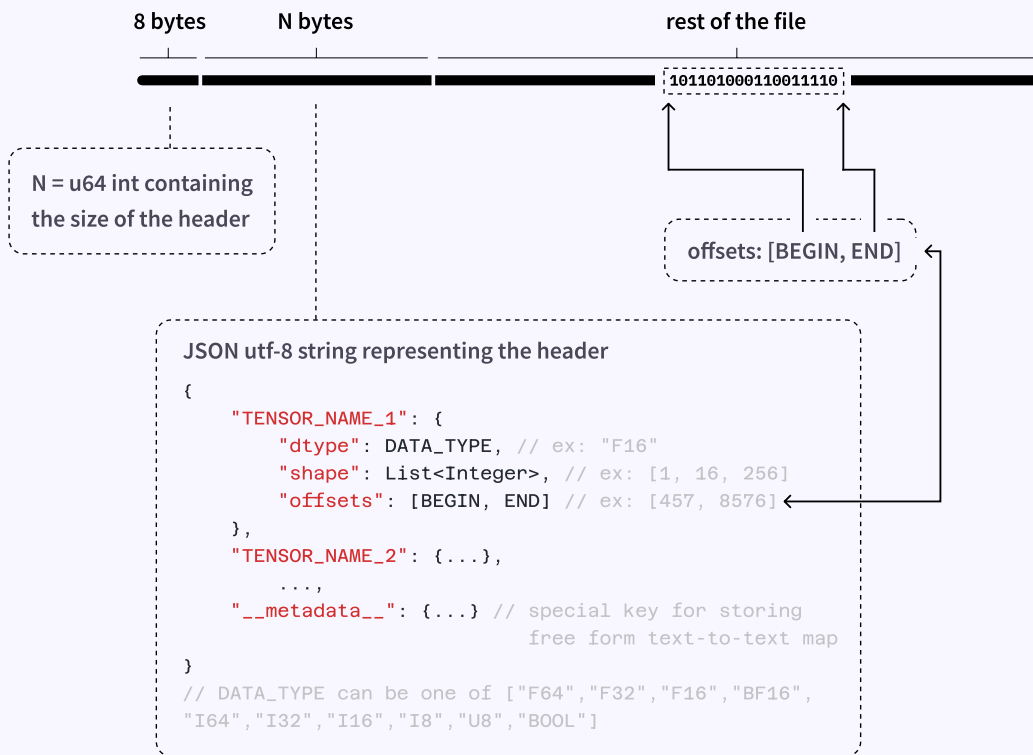
Save tensors

```
import torch
from safetensors.torch import save_file

tensors = {
    "embedding": torch.zeros((2, 2)),
    "attention": torch.zeros((2, 3))
}
save_file(tensors, "model.safetensors")
```

Format

Let's say you have safetensors file named `model.safetensors`, then `model.safetensors` will have the following internal format:

File name **model.safetensors**

Featured Projects

Safetensors is being used widely at leading AI enterprises, such as [Hugging Face](#), [EleutherAI](#), and [StabilityAI](#). Here is a non-exhaustive list of projects that are using safetensors:

- [huggingface/transformers](#)
- [ml-explore/mlx](#)
- [huggingface/candle](#)
- [AUTOMATIC1111/stable-diffusion-webui](#)
- [Llama-cpp](#)
- [microsoft/TaskMatrix](#)
- [hpcaitech/ColossalAI](#)
- [huggingface/pytorch-image-models](#)
- [CivitAI](#)

- [huggingface/diffusers](#)
- [coreylowman/dfdx](#)
- [invoke-ai/InvokeAI](#)
- [oobabooga/text-generation-webui](#)
- [Sanster/lama-cleaner](#)
- [PaddlePaddle/PaddleNLP](#)
- [AIGC-Audio/AudioGPT](#)
- [brycedrennan/imaginAIry](#)
- [comfyanonymous/ComfyUI](#)
- [LianjiaTech/BELLE](#)
- [alvarobartt/safejax](#)
- [MaartenGr/BERTopic](#)
- [rachthree/safestructures](#)
- [justinchuby/onnx-safetensors](#)

</> [Update](#) on GitHub

Speed Comparison →